Ternary Conversion and Mixing of Open-Weight Models via Hodge–Kuramoto Alignment

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Abstract

We convert published open-weight checkpoints to the ternary $(-1, 0, +1)$ format of Microsoft BitNet b1.58 2B4T [1] and mix them into it. Each weight matrix is a 1-cochain on the bipartite graph of its input and output channels. A mixed-radix factorization of each channel length orders channels into a multi-radix heap (MRH) keyed by ternary conversion error; a two-way (down/up) Laplacian is diagonalized by the mixed-radix DFT, which yields an exact spectral Hodge decomposition into gradient, curl and harmonic (low-bin) parts. Phase correlation per digit and Re-Basin weight matching [3] seed the alignment; a discrete Kuramoto lock with the gains of the author's `jev_kuramoto` oscillator engine aligns the harmonic modes; ternary-native TIES [4] resolves the curl part. Whitney prolongation and de Rham restriction [9] resize between widths, a randomized Hadamard residual rotation [5, 6] precedes absmean ternarization [2], and a signed, invertible BNZ4 channel-map chunk records the mapping. Every constant is derived in Appendix A. We report only operator checks and tiny synthetic runs; large-model evaluation is given as a protocol.

1 Setting and decisions

The base is BitNet b1.58 2B4T [1]: every attention and feed-forward linear layer is ternary with one absmean scale, stored in the Champ packed format (four trits per byte plus a BF16 `weight_scale`); embeddings and norms stay in BF16. Other open checkpoints (FP16/BF16/F32) are converted to that format and mixed into it. The pipeline order is fixed: align → resize → mix in full precision → ternarize (rotation + absmean) → distill into 2B4T and its adapters. Models that are already ternary on the same base (BitNet fine-tunes, Champ adapters) skip alignment and use ternary-native TIES directly. Teachers for distillation supply top-k logits. Distillation is outside the code described here (Section 9).

2 Channel complex and Hodge decomposition

Complex. For a weight $W \in \mathbb{R}^{R\times C}$ the nodes of K are the C input and R output channels (or their MRH buckets), the edges are the entries (r, c), and the 2-cells are the elementary squares of the product of the two channel cycles. The channel map is the 1-cochain

$$M(r, c) = W_{rc}/\gamma_W, \qquad \gamma_W = \tfrac{1}{RC}\sum |W_{rc}|. \tag{1}$$

Hodge decomposition [7, 8]. With coboundaries d_0, d_1 and Hodge stars $\star_k$,

$$M = d_0\alpha \;\oplus\; \delta_1\beta \;\oplus\; h, \qquad \delta_1 = \star_0^{-1} d_0^{\top} \star_1, \tag{2}$$

$$L_1 = d_0 \star_0^{-1} d_0^\top \star_1 + \star_1^{-1} d_1^\top \star_2 d_1, \qquad h \in \ker L_1. \tag{3}$$

With unit stars (DECLUT: $\star_1 = I$, $\delta_1 = -d_0^\top$ up to orientation) the three parts are orthogonal. The gradient part is global reorder and scale (absorbed by alignment); the curl part is merge conflict (resolved by ternary TIES); the harmonic part is shared structure (preserved under resizing and aligned by the Kuramoto lock).

Two-way Laplacian in the mixed-radix DFT domain. The down Laplacian acts on the source (input-channel) side, the up Laplacian on the target (output-channel) side. Order each side by mixed-radix digits $N = r_1 r_2 \cdots r_m$ with every digit level circulant; the mixed-radix DFT [11] diagonalizes the Laplacian:

$$\lambda_k = \sum_j \Big(2 - 2\cos\frac{2\pi k_j}{r_j}\Big), \qquad k = (k^{\text{row}}, k^{\text{col}}). \tag{4}$$

The spectral projectors are indicator sets on bins:

$$\begin{aligned} &\mathcal{G} = \{k : k^{\text{row}} = 0 \text{ or } k^{\text{col}} = 0\}, \qquad \mathcal{C} = \text{all other bins}, \\ &\mathcal{H} = \{k \notin \mathcal{G} : \text{ each side has exactly one digit } k_j \neq 0, \text{ and it is } k_j \in \{1, r_j - 1\}\}. \end{aligned} \tag{5}$$

$\mathcal{G}$ is exactly the image of the row/column potentials ($d_0\alpha$); $\mathcal{H}$ is "the low bins of each digit": the lowest nonzero bin of one digit on each side ($\lambda_k = \lambda_{\text{row}}^{\min} + \lambda_{\text{col}}^{\min}$ per digit pair); $\mathcal{C}$ carries the remaining rotational energy. By Parseval the three parts are orthogonal (Test S3).

3 Mixed-radix index and multi-radix heap

Index length. Factor a channel length into primes, coarsest digit (largest prime) first, e.g. $2560 = 5 \times 2^9$, $6912 = 3^3 \times 2^8$. The index keeps whole radix levels:

$$G(n) = \prod_{j \le J} r_j, \qquad J = \max\{J : \textstyle\prod_{j \le J} r_j \le 2^{2b}\},\ b = 4, \tag{6}$$

so $G(2560) = 160$, $G(6912) = 216$. $2^{2b} = 256$ is two MRH levels of 16 buckets. Resizing keeps or drops DFT bins per radix level; only the low bins $\mathcal{H}$ are stored.

MRH key. For each output (input) channel c, with $\text{RoundClip}(x) = \max(-1, \min(1, \text{round}(x)))$,

$$\kappa_c = \min\Big(2^{12} - 1,\ \Big\lfloor 2^{12}\, \frac{\frac{1}{m}\sum_j |W_{cj} - \gamma\, \text{RoundClip}(W_{cj}/\gamma)|}{\gamma} \Big\rfloor\Big), \tag{7}$$

a 12-bit saturating integer key (as in the DMT chip arrival field), $\gamma = \gamma_W$, $\kappa_c = 0$ if $\gamma = 0$. Channels with the smallest conversion error pop first and are therefore matched, merged or split first.

Channel coordinate (spline knots). The MRH pop order, ties broken FIFO by index, gives

$$x_c = (\text{rank}_{(\kappa,c)} + \tfrac{1}{2})/n. \tag{8}$$

Bucket. A tensor's MRH bucket is the top 4-bit digit of its median key:

$$b_W = \lfloor \text{median}_c\, \kappa_c / 2^8 \rfloor \in \{0, \ldots, 15\}. \tag{9}$$

4 Resizing: Whitney prolongation and de Rham restriction

On each side the index is a cycle of G cell-centred nodes. A source channel map is carried onto the index by the degree-1 spline over x_c, which is the Whitney 0-form (hat-function) interpolant followed by the de Rham map (evaluation at index nodes) [9, 10]:

$$\widehat{M} = \mathcal{R}_G \, \mathcal{P}_x \, M, \qquad (\mathcal{P}_x f)(y) = \sum_c f_c \, \phi_c(y), \;\; \phi_c \text{ hat functions on } \{x_c\}. \tag{10}$$

The same operator (with the roles reversed) samples the index at the target channels; its per-channel weights $(\ell, 1-t, t)$ are stored in the BNZ4 chunk. Depth (layers) uses the same hat on normalized depth:

$$w_s(u_t) = \omega_s \max\bigl(0, 1 - |u_s - u_t|(L_s - 1)\bigr), \qquad u = \ell/(L-1), \tag{11}$$

with ω_s the user mixing weight of model s and L_s its layer count ($w_s = \omega_s$ if $L_s = 1$). For nested cycles the operators are, per digit axis,

$$\begin{aligned} &(d_0 a)_e = a_{e+1} - a_e, && P_0 a: \;\; (P_0 a)_{2i} = a_i, \;\; (P_0 a)_{2i+1} = \tfrac{1}{2}(a_i + a_{i+1}), \\ &P_1 b: \;\; (P_1 b)_{2e} = (P_1 b)_{2e+1} = \tfrac{1}{2} b_e, && R_0: \;\; (R_0 a)_i = a_{2i}, \quad R_1: \;\; (R_1 b)_e = b_{2e} + b_{2e+1}, \end{aligned} \tag{26}$$

which satisfy $d_0 P_0 = P_1 d_0$, $R_0 P_0 = I$, $R_1 P_1 = I$, $d_0 R_0 = R_1 d_0$, and map the harmonic 1-cochain of the cycle to a harmonic cochain ($L_1 P_1 h = 0$), so cohomology is preserved (Test S5). (Equation (26) is numbered to match the code.)

5 Alignment tuning algorithm

Seed shifts (phase correlation per digit) [12]. For grids A (contributor) and B (reference = base):

$$X = \frac{\widehat{B}\,\overline{\widehat{A}}}{|\widehat{B}\,\overline{\widehat{A}}|} \;\; (X = 0 \text{ where the product is 0}), \qquad s_j = \arg\max_t \sum_{k:\, k_j = t} \mathcal{F}^{-1} X, \tag{12}$$

applied digit by digit (coarse to fine), as in the author's FFT correction chip.

Seed permutation (Re-Basin weight matching). Rows then columns solve

$$\pi = \arg\max_{\pi \in S_G} \; \langle B, \, P_\pi A \rangle_F, \tag{13}$$

a linear assignment problem (Hungarian method). This is Git Re-Basin weight matching [3]; exact optimal transport with uniform marginals and the same cost has the same optimum (Birkhoff–von Neumann: the LP optimum is attained at a permutation), so "Re-Basin or OT" coincide here.

Kuramoto lock on harmonics. Each pair (model i, harmonic bin $m \in \mathcal{H}$) is an oscillator with phase $\theta = \arg \widehat{M}_i(m) - \arg \widehat{M}_{\text{base}}(m)$ evolving by

$$\dot{\theta}_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i), \tag{14}$$

$$\omega_i = 0. \tag{14b}$$

All oscillators of bin m share eigenvalue λ_m, hence the same natural frequency; in the frame co-rotating at that frequency $\omega_i = 0$. Time is discretized as one tick of the `jev_kuramoto` engine (R-KTE-1, R-KTE-8: the neighbour term is the DEC Laplacian $-\delta_1 d_0 \theta$ scaled by 2^{-g_θ}):

$$\theta_i \leftarrow \theta_i + \sum_j K_{ij} \sin(\theta_j - \theta_i), \qquad K_{ij} = 2^{-g_\theta} \frac{c_{ij}}{d_{\max}}, \quad c_{ij} = \begin{cases} 1 & \text{same MRH bucket or digit level} \\ 2^{-(g_\omega - g_\theta)} & \text{otherwise} \end{cases} \tag{15}$$

with $g_\theta = 4$, $g_\omega = 8$, $d_{\max} = \max_i \sum_j c_{ij}$. By Gershgorin, $\lambda_{\max}(L_K) \le 2 \cdot 2^{-g_\theta} < 2$, so the explicit tick is stable. The heap-weight coupling is thus: strong (2^{-4}) within an MRH bucket or digit level, weak (2^{-8}) across.

$$\text{locked}_k \iff |\operatorname{wrap}(\theta_k - \theta_{\text{ref}(k)})| \le W_{\text{lock}} = \tfrac{2\pi}{16}, \qquad \text{lost after } N_{\text{lost}} = 255 \text{ ticks.} \tag{16}$$

$$r = \left| \tfrac{1}{n} \textstyle\sum_k e^{i\theta_k} \right|, \qquad r_m = \left| \tfrac{1}{|S|} \textstyle\sum_i e^{i\theta_{i,m}} \right|. \tag{17}$$

Anneal (coarse to fine, MRH-weighted). Stage $s = 0, \ldots, D-1$ activates oscillators whose bin's finest nonzero digit is $\le s$ and ramps the coupling $K \cdot q / N_{\text{lost}}$, $q = 1..N_{\text{lost}}$; refine then runs windows of N_{lost} ticks at full K, at most $D+2$ windows:

$$T_{\text{anneal}} = D\, N_{\text{lost}}, \qquad T_{\text{refine}} \le (D+2) N_{\text{lost}}. \tag{18}$$

Merge. For a bin whose base oscillator is locked, with locked set S_m (members locked to the base), weights w_i from (11) and window-averaged lag $\bar{\phi}_i = \arg\langle e^{i(\theta_i - \theta_{\text{base}})} \rangle$:

$$\widehat{H}(m) = (1 - r_m) \widehat{H}_{\text{base}}(m) + r_m \frac{\sum_{i \in S_m} w_i |\widehat{M}_i(m)| e^{i(\arg \widehat{M}_{\text{base}}(m) + \bar{\phi}_i)}}{\sum_{i \in S_m} w_i}; \tag{19}$$

unlocked bins stay with the base (or route to adapters). **Refine stop.** Re-project onto $\mathcal{H}$ each pass; stop when

$$\|\widehat{H}^{(p)} - \widehat{H}^{(p-1)}\| / \|\widehat{H}^{(p-1)}\| < 2^{-8} \ \wedge \ |r^{(p)} - r^{(p-1)}| < 2^{-8}, \qquad p \le D + 1. \tag{20}$$

Ternary TIES on the curl part [4]. Deltas $\Delta_i = \Pi_{\mathcal{C}}(M_i - M_{\text{base}})$ are coded $t_i = \text{RoundClip}(\Delta_i / \gamma_{\Delta_i})$; the elected sign is $\sigma = \operatorname{sgn} \sum_i w_i t_i$ and

$$\Delta^\star = \frac{\sum_{i : t_i = \sigma \neq 0} w_i \Delta_i}{\sum_{i : t_i = \sigma \neq 0} w_i}. \tag{21}$$

Synthesis and resize.

$$\begin{aligned} W_t = \bar{\gamma} \Big[& \mathcal{R}_t \mathcal{P} \big(\Pi_{\mathcal{G}} M_{\text{base}} + \Pi_{\mathcal{C}} M_{\text{base}} + \Delta^\star + \mathcal{F}^{-1} \widehat{H} \big) \\ & + (I - \mathcal{R}_t \mathcal{P}\, \mathcal{R}_G \mathcal{P}_x) M_{\text{base}} \Big], \qquad \bar{\gamma} = \frac{\sum w_s \gamma_s}{\sum w_s}, \end{aligned} \tag{22}$$

the last term (base residual, the two-grid complement) present only when the base has the target shape.

Algorithm 1 (Hodge–Kuramoto alignment tuning, per target tensor). Input: contributors M_i on the index grid (10), weights w_i (11), buckets (9).

1. **Seed:** phase-correlate each M_i to the base digit by digit (12); Re-Basin rows then columns (13).
2. Hodge-split each contributor in the DFT domain (4)–(5); ternary TIES on curl deltas (21).
3. **Anneal:** oscillators on $\mathcal{H}$ with coupling (15), stages coarse→fine digits (18).
4. **Refine:** for $p = 1, \ldots, D+1$: tick windows of N_{lost} (15)–(16), merge (19), re-project on $\mathcal{H}$; stop at (20).
5. Synthesize and resample at the target channels (22); rotate (24); ternarize (25).
6. **Emit:** the BNZ4 channel-map chunk (Section 7).

6 Ternarization: rotation plus absmean

Randomized Hadamard rotation [5]. For hidden size H let b be the largest power of two dividing H ($b = 512$ for $H = 2560$) and

$$Q = D\,(I_{H/b} \otimes H_b)/\sqrt{b}, \qquad D = \operatorname{diag}(\pm 1) \text{ Rademacher}, \tag{23}$$

orthogonal for any signs. Rotating the residual stream $x \mapsto Q^\top x$ leaves an RMSNorm transformer's function unchanged once each RMSNorm gain g is folded into the readers that follow it (computational invariance [6, 5]):

$$\begin{aligned} &\text{readers } (q,k,v,\text{gate},\text{up}):\ W \leftarrow W\operatorname{diag}(g)Q, \quad \text{writers } (o,\text{down}):\ W \leftarrow Q^\top W, \\ &\text{embedding } E \leftarrow EQ, \qquad\qquad\qquad\quad \text{head } W_{\text{lm}} \leftarrow W_{\text{lm}}\operatorname{diag}(g_{\text{final}})Q, \quad g \leftarrow \mathbf{1}. \end{aligned} \tag{24}$$

BitNet's inner sub-norms act on the attention/FFN inner activations, not on the residual stream, and are untouched. A tied head is written untied. If any residual reader or writer lacks a recognized RMSNorm, rotation is not applied (logged), because invariance could not be guaranteed. **Absmean** [2].

$$\gamma = \tfrac{1}{RC}\textstyle\sum |W|, \qquad \widetilde{W} = \operatorname{RoundClip}(W/\gamma), \qquad \widetilde{W} = 0 \text{ if } \gamma = 0. \tag{25}$$

7 BNZ4 signed, invertible channel-map chunk

Stored per tensor as `bnzmix.bnz4.<name>` (U8) beside the packed trits and `weight_scale`; little-endian; a loader that reads it reproduces the mapping exactly.

field	type	meaning
magic, version	`"BNZ4"`, u16=1	
name	u16 length + UTF-8	tensor name
$R_{\text{out}}, C_{\text{out}}, G_r, G_c$	4×u32	output shape, index grid (6)
n_r, n_c, radices	u8,u8, $(n_r{+}n_c)$×u8	digits, coarsest first
S	u16	contributors; each: src u16, bucket u8 (9), weight f32 (11), digit shifts $(n_r{+}n_c)$×u8 (12), signed perms $G_r{+}G_c$ i32 (13)
row/col prolongation	$(R_{\text{out}}{+}C_{\text{out}})$×(u32,f32,f32)	left index node, hat weights (10)
low bins	u32 B; B×(u32,f32,f32)	bin index, Re, Im of $\widehat{H}$ on $\mathcal{H}$
CRC-32	u32	IEEE CRC of all preceding bytes

Permutation entries are $\pm(p+1)$: the sign is a channel sign flip, the magnitude the 1-based target index. The decoder rejects any entry set that is not a bijection, so every stored map is invertible (inverse: p^{-1} with the same signs). The chunk travels inside the lossless BNZ container.

8 Measured results (tiny synthetic tests only)

All numbers are from the self-test and from tiny synthetic checkpoints (hidden size 16–32, 2–3 layers) in `ternmix-py/test`; no real or 2B-scale model was run.

check	result
S1 λ_k vs dense Laplacian, $n \in \{12, 16, 30, 60, 64\}$	$\leq 2 \times 10^{-14}$
S2 explicit-complex Hodge (random stars): $\|L_1 h\|$, cross terms	5.7×10^{-14}, $\leq 2 \times 10^{-17}$
S3 spectral Hodge recon / cross terms (32×30)	1.0×10^{-15} / $\leq 9 \times 10^{-18}$
S4 Hadamard max $\|QQ^\top - I\|$, $H \in \{32, 40, 48, 2560\}$	$\leq 1.1 \times 10^{-16}$
S4 rotation invariance (RMSNorm reader, float32 weights)	1.9×10^{-7}
S5 $\|d_0 P_0 - P_1 d_0\|$, $\|R_0 P_0 - I\|$, $\|R_1 P_1 - I\|$, $\|L_1 P_1 h\|$	1.1×10^{-16}, 0, 0, 0
S6 Kuramoto tick, 24 osc., spread $\pm\pi/4$ / $\pm 0.95\pi$	$\langle r \rangle = 0.99993$ / 0.99965; 24/24 locked
S7 BNZ4 round trip / CRC / non-bijective perm	exact / rejected / rejected
S8 seed: phase correlation and Hungarian on known roll/perm	exact (0 error)
tiny run, 3 sources, target config (21 tensors)	all lock passes converged, $\langle r \rangle = 1.0$, locked = n_{osc}
tiny run BNZ round trip	byte-identical

With $\omega_i = 0$ and connected coupling, phase synchronization is the generic outcome [13, 14], which is what the tiny runs show; the lock's informative output is the consensus lag $\bar{\phi}_i$ and the lost set, which stays empty unless contributors sit near anti-phase. Whether a non-zero, data-derived ω improves merges is an open question for the protocol below.

9 Measurement protocol (future work)

No large-model numbers are claimed. To be measured on BitNet b1.58 2B4T plus open checkpoints: perplexity and KL to the full-precision mix after ternarization, with and without rotation (24); merge KL with and without alignment (12)–(19); Hodge energy split $(\mathcal{G}, \mathcal{C}, \mathcal{H})$ against merge KL; r as a predictor of merge success; low-bin DFT energy fraction after MRH channel sorting (8); BNZ4 chunk size against a dense permutation; distillation into 2B4T and its adapters from top-k teacher logits.

A Parameter derivations

term	value	derivation / source
roles	q,k,v,o,gate,up,down,lm_head	BitNet replaces every attention/FFN linear with BitLinear; embeddings and head stay high precision [1].
index length (6)	≤ 256, whole radix levels	two MRH levels of 16 buckets (DMT three-level 16-bucket ordering unit); resizing by radix level (decision 4).
spline degree / knots (8),(10)	degree 1, knots = MRH pop order	Whitney 0-forms are the degree-1 hats [9]; knots come from the heap order, no count or jitter needed.

term	value	derivation / source
heap key (7)	12-bit, saturating	DMT chip 12-bit arrival field saturating at 4095.
bucket (9), levels	16 buckets, top 4-bit digit	DMT ordering unit: 16 buckets per level.
depth weight (11)	Whitney hat	same prolongation as (10) on the layer axis; replaces Gaussian width/cutoff.
ω_i (14b)	0	same eigenvalue per bin $\Rightarrow$ equal natural frequencies; co-rotating frame.
strong / weak K (15)	2^{-4} / 2^{-8}	`jev_kuramoto` gains $g_\theta = 4$, $g_\omega = 8$ (R-KTE-1, R-KTE-8).
normalization $d_{\max}$ (15)	$\max_i \sum_j c_{ij}$	Gershgorin stability of the explicit tick.
time step	1 tick	KTE is a per-tick update (R-KTE-1).
lock window (16)	1/16 turn	KTE W_{lock} (R-KTE-2).
lost / stage / window (16),(18)	255 ticks	KTE N_{lost}.
refine tol (20)	2^{-8}	BF16 has 8 significant bits: smaller changes are not representable in the stored model.
max passes (20)	$D + 1$	one pass per digit level plus the anneal pass.
harmonic set (5)	lowest bin per digit	"harmonic and smooth modes occupy the low bins of each digit" (v1).
seed assignment (13)	Re-Basin weight matching	OT with uniform marginals has the same optimum (Birkhoff).
phase-corr./absmean ε	none	division guarded exactly: $X = 0$ where $\lvert\cdot\rvert = 0$, $\widetilde{W} = 0$ if $\gamma = 0$.
rotation (23)	randomized Hadamard	QuaRot [5]; invariance by gain folding [6]. Seed only fixes D; any D is valid.
base residual (22)	on	two-grid complement keeps detail finer than the index.
float dtype	BF16	BitNet 2B4T stores non-ternary tensors in BF16 [1].
stored bins	$\lvert\mathcal{H}\rvert$	"store only low-frequency coefficients" (decision 4).
triangles	grid squares	data-free: 2-cells of the product channel cycles; co-activation triangles need calibration data (open).

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